

CSC490 A3

Benson Li 1007815376
Yuchen Wang 1010441595
Xuanyi Lyu 1009343266
Zheyu Zheng 1008979580

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Part One: Diagramming Your Infrastructure

The following diagram provides a detailed overview of our project's current infrastructure.

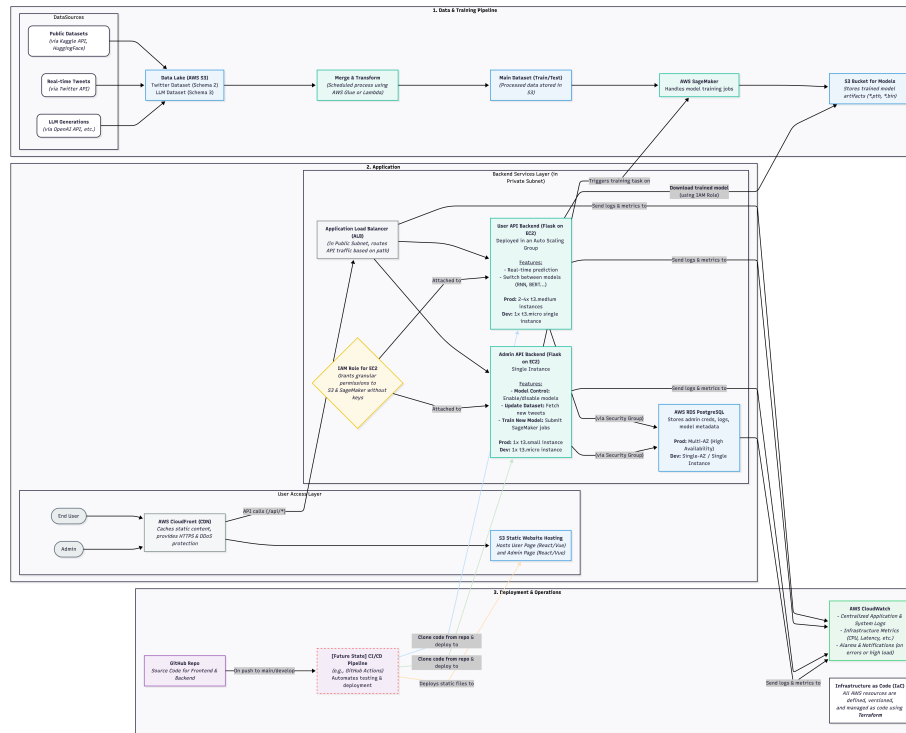


Figure 1: High-level infrastructure diagram for the TweetVerify project.

A high-resolution version of the diagram is also available online at: [Mermaid Chart Link](#).

Differences Between Our Development and Production Environments

1. In the **development environment**, we use a single small EC2 instance (e.g., `t3.micro`), while the **production environment** runs on larger instances (e.g., `t3.medium` or higher) with production-grade settings and monitoring.
2. **Backups and encryption** are enabled for the RDS database in the production environment but disabled in the development environment.
3. The development environment uses **local logs or console outputs** for debugging, whereas the production environment is integrated with **Amazon CloudWatch** for centralized logging, error tracking, and metrics.
4. In development, **Flask debug mode** is enabled with minimal instance resources, while in production, **debug mode is disabled** and the application is served using **Nginx** for optimized performance.

Part Two: Infrastructure as Code Implementation

We chose Terraform as our Infrastructure as Code tool. The implementation code is available at: [Our GitHub Repository](#).

Part Three: Disaster Recovery Demonstration

A video demonstrating our disaster recovery process can be found at the following link: [Watch the Video](#)

(The pre-trained models are only used for demonstration purposes; the frontend and backend have been newly developed and conform to all CSC490 specifications.)